Lesson 1: Introduction to Tide Pools

Lesson Plan Topic: Introduction to Tide Pools
Instructor: Tadeh Peroomian
Student/Audience Description: Friends

Amount of Time for the Lesson: 5 minute introduction, 20 minute lecture, 20 minute exploration, 10 minute reflection, 5 minute conclusion. 1 hour class total.

Management Considerations(optional):

Materials: Notes, Containers, Phones with Seek by iNaturalist, Merlin

Curriculum Goal: I want them to leave this lesson more knowledgeable about the tide pools.

Cognitive Objectives:

1. The students will be able to compare and contrast what they saw in the tide pools based on the tides.
2. Using iNaturalist, the students will be able to identify at least 5 of the species present at the tide pools.

Affective Objectives:

1. The students will feel motivated to visit the tide pools again.
2. The students will be curious about the things they see in the tide pools.

Step-by-Step Procedures: Using Natural Learning Cycles:

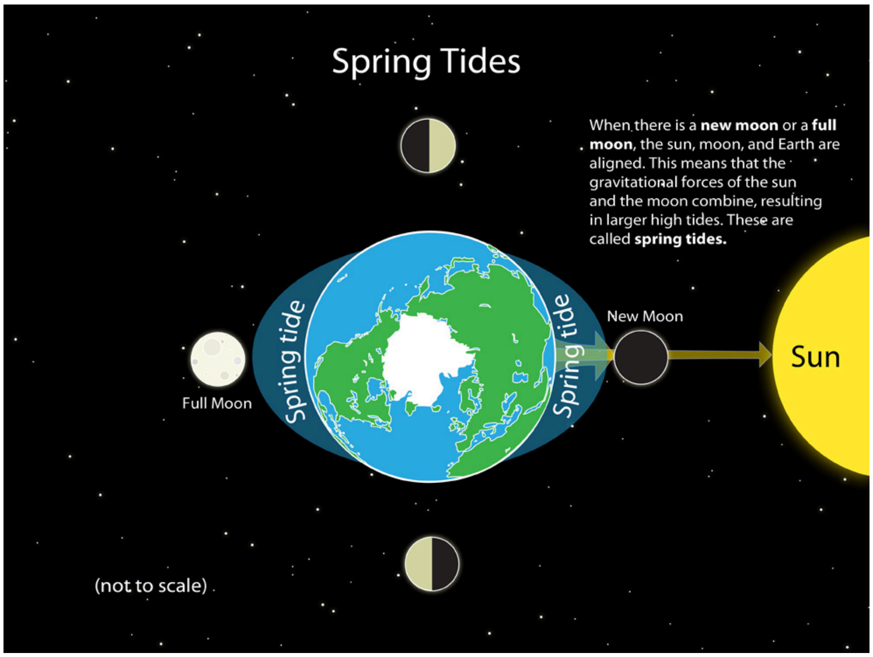
East: Tell them a personal story to draw their interest. Mine is about when I first went tide pooling in Baja California on a study abroad trip, and how that was an experience that brought back my desire to study marine biology and environments. I wanted to share with them something that I love, and hope that they enjoy it. Ask them if they have any stories about tide pools to share.

Southeast: Share the rules about how to interact with tide pools in a safe way for both them and the environment. This includes watching where you step to not crush any organisms, not flipping over rocks in order to not disturb the environment, and handling animals with care.

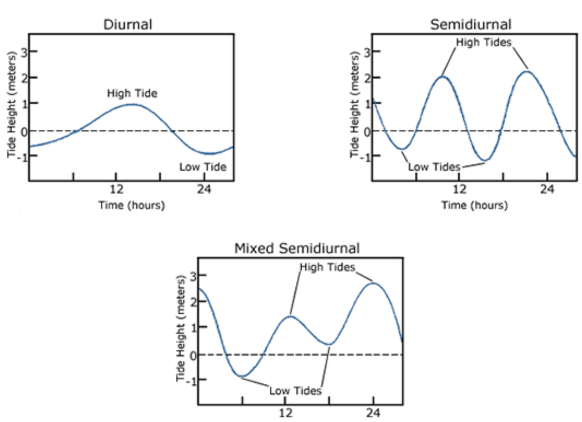
South: Teaching a lesson on tide pools. The main topics of the lesson will be: what is a tide pool, how they are formed, and how the tidal calendar and tides relate to it.

Tide pools are isolated pockets of seawater found within the intertidal zone. They are found on rocky coasts, which have depressions to trap water as the tide progresses and retreats. Many of these pools exist as separate bodies of water only at low tide, as seawater gets trapped when the tide recedes.

Tides are caused mainly by the gravitational pull of the moon and the sun on Earth's oceans. The moon's gravity creates bulges of water that produce high tides, while areas between the bulges experience low tides as Earth rotates. The sun also affects tides, making them stronger or weaker depending on the positions of the sun, moon, and Earth.



A tidal cycle is usually about 25 hours. Because the Earth is not a perfect sphere, the tidal cycle varies by location. There are three tidal cycle types.



The Diurnal tide cycle experiences one high and one low tide every lunar day. Many areas in the Gulf of Mexico experience these types of tides.

The Semidiurnal tide cycle experiences two high and two low tides of approximately **equal** size every lunar day. Many areas on the eastern coast of North America experience these tidal cycles.

The Mixed Semidiurnal tide cycle experiences two high and two low tides of **different** sizes every lunar day. Many areas on the western coast of North America experience these tidal cycles.

Have students look at the tidal calendar, and ask them which tidal cycle they think we experience in Southern California/Isla Vista.

Southwest: Students will be allowed to explore the tide pools on their own for 5 minutes.

South/Southwest: Next, ask the students to circle up around the most interesting thing they saw in the tide pools. Have them open the iNaturalist app, and show them a quick demonstration of how it works.

Students will be allowed to explore the tide pools, using iNaturalist to observe and identify the organisms that they find.

South/Southwest: Do the same for the Merlin app to identify shorebirds.

West: After they have had some time to walk around, gather them together and have them share with the class what they found and why they found it interesting.

Northwest: Have them reflect on what they learned, what they saw, what they will take with them from this lesson, and what surprised them. How did this lesson make them feel? Ask them, "Does this motivate you to bring other people to the tide pools? Why or why not?" "What are you still curious about?"

North: Encourage them to use iNaturalist in other ecosystems that they are curious about.

Northeast: Finally, leave them with an invitation to come back to enjoy the tide pools with others. Tell them about the next time we will meet.

Background Information

4MAT:
Type 1: Type 1 learners favor experiencing and reflecting. During this lesson, they will be able to experience exploring the tide pools, and also have time to reflect on what they learned at the end of the lesson.
Type 2: Type 2 learners favor thinking and reflecting. During this lesson, they will be able to think critically about the information presented in the lecture, and have an opportunity to reflect at the end of the lesson.
Type 3: Type 3 learners favor thinking and acting. During this lesson they will be able to actively explore and learn from the tide pools, while thinking about the topics discussed in the lecture.
Type 4: Type 4 learners favor experiencing and acting. During this lesson they will be able to actively explore and experience the tide pools.

Lesson 2: Exploring the UCSB

Lesson Plan Topic: Exploring the UCSB Lagoon
Instructor: Tadeh Peroomian
Student/Audience Description: Friends

Amount of Time for the Lesson: 5 minute introduction, 20 minute lecture, 20 minute exploration, 10 minute reflection, 5 minute conclusion. 1 hour class total.

Management Considerations(optional):

Materials: Notes, Phones with Seek by iNaturalist, Merlin

Curriculum Goal: I want them to leave this lesson more knowledgeable about the lagoon.

Cognitive Objectives:

1. The students will be able to compare and contrast what they saw in the tide pools based on the tides.
2. Using iNaturalist, the students will be able to identify at least 5 of the species present at the tide pools.

Affective Objectives:

1. The students will feel motivated to visit the lagoon again.
2. The students will be curious about the things they see around the lagoon.

Step-by-Step Procedures: Using Natural Learning Cycles:

East: We're getting into finals season, so I want to say that I am thankful for them coming out here for me, and start a gratitude circle.

Southeast: Tell them the plan for the day of the short lecture before sending them off to explore the lagoon and its various organisms.

South:
The UCSB Lagoon was originally a natural coastal wetland that supported diverse wildlife and was part of the traditional homeland of the Chumash people. When UCSB moved to its current site in the 1950s, the area was modified during campus construction, creating the lagoon's present shape.

In recent decades, restoration efforts led by the Cheadle Center for Biodiversity and Ecological Restoration have improved water quality and restored native habitats, making the lagoon an important ecological, research, and educational resource.

The UCSB Lagoon supports a diverse community of plants and animals adapted to coastal wetland environments. Common plants include pickleweed, cattails, and native coastal sage scrub species, which provide food and shelter for wildlife.

The lagoon is home to many bird species, such as Great Egrets, Great Blue Herons, Brown Pelicans, and Mallards, while its waters support fish like the California killifish and longjaw mudsucker. Other animals commonly found around the lagoon include insects, amphibians, and small mammals, all of which contribute to the ecosystem's biodiversity.

Southwest: Students will be allowed to explore the lagoon on their own, utilizing iNaturalist and Merlin to identify various organisms for 10 minutes.

West: After they have had some time to walk around, gather them together and have them share with the class what they found and why they found it interesting.

Northwest: Have them reflect on what they learned, what they saw, what they will take with them from this lesson, and what surprised them. How did this lesson make them feel? Ask them, “What did you notice here that was different from the tide pools, another habitat a couple hundred feet away?” “What are you still curious about?”

North: Encourage them to use iNaturalist in other ecosystems that they are curious about, or just in their day-to-day life.

Northeast: Finally, tell them about the next time we will meet, at the REEF.

Background Information

4MAT:
Type 1: Type 1 learners favor experiencing and reflecting. During this lesson, they will be able to experience exploring the lagoon, and also have time to reflect on what they learned at the end of the lesson.
Type 2: Type 2 learners favor thinking and reflecting. During this lesson, they will be able to think critically about the information presented in the lecture, and have an opportunity to reflect at the end of the lesson.
Type 3: Type 3 learners favor thinking and acting. During this lesson they will be able to actively explore and learn from the lagoon, while thinking about the topics discussed in the lecture.
Type 4: Type 4 learners favor experiencing and acting. During this lesson they will be able to actively explore and experience the lagoon.

Lesson 3: Trip to the REEF

Lesson Plan Topic: Trip to the REEF
Instructor: Tadeh Peroomian
Student/Audience Description: Friends

Amount of Time for the Lesson: 5 minute introduction, 20 minute lecture, 20 minute exploration, 10 minute reflection, 5 minute conclusion. 1 hour class total.

Management Considerations(optional):

Materials: Notes

Curriculum Goal: I want them to leave this lesson more knowledgeable about the tide pools.

Cognitive Objectives: The students will be able to compare and contrast what they saw in the tide pools at each beach.

Affective Objectives: I want them to feel welcome to visit the tide pools again.

Step-by-Step Procedures: Using Natural Learning Cycles:

East: Go around in a circle and ask what tide pool species they are more interested in seeing.

Southeast: Tell them the plan for the day of letting them roam the REEF. Tell them that you will also be walking around, and that they can ask you or one of the employees if they have any questions about the species.

South/Southwest: Let the students explore the REEF with Seek by iNaturalist, and be prepared to share any information they would like to know, including some of the information below:

Lobster (Arthropods): Lobsters are arthropods with jointed legs, segmented bodies, and a hard exoskeleton. They molt their shells as they grow.

Moray Eel: Moray eels are long, predatory fish that live in reefs and rocky crevices. They use sharp teeth to catch fish and crustaceans.

Echinoderms: Echinoderms are marine animals with spiny skin, radial symmetry, and tube feet used for movement and feeding.

Sea Stars (Pisaster): Sea stars are echinoderms that use tube feet to move and feed. Pisaster ochraceus is a keystone predator that helps maintain ecosystem balance.

Sea Cucumber: Sea cucumbers are soft-bodied echinoderms that feed on organic matter in seafloor sediments.

Sea Urchins: Sea urchins are spiny echinoderms that graze on algae using a specialized mouth called Aristotle's lantern.

Swell Shark Eggs: The Swell shark lays tough egg cases ("mermaid's purses") that protect developing embryos.

Sea Star Wasting Disease: A disease that causes sea stars to develop lesions, lose arms, and die, leading to major population declines.

West: After they have had some time to walk around, gather them together and have them share with the class what they found and why they found it interesting.

Northwest: Have them reflect on what they learned, what they saw, what they will take with them from this lesson, and what surprised them. How did this lesson make them feel? Ask them, “Now that you have had a couple classes using it, how do you feel about the Seek by iNaturalist and Merlin Bird ID apps?” “What are you still curious about from today?”

North: Thank them for their time and share the feedback link before saying goodbye.

4MAT:
Type 1: Type 1 learners favor experiencing and reflecting. During this lesson, they will be able to experience exploring the tide pools, and also have time to reflect on what they learned at the end of the lesson.

Type 2: Type 2 learners favor thinking and reflecting. During this lesson, they will be able to think critically about the information presented by the employee or me, and have an opportunity to reflect at the end of the lesson.

Type 3: Type 3 learners favor thinking and acting. During this lesson they will be able to actively explore and learn from the tide pool touch tanks, while thinking about the topics discussed in the lecture.

Type 4: Type 4 learners favor experiencing and acting. During this lesson they will be able to actively explore and experience the tide pool touch tanks.